

Tensor Crib Sheet

$$\lambda^{a'} = X_{b'}^{a'} \lambda^b \quad \text{where} \quad X_{b'}^{a'} = \frac{\partial x^{a'}}{\partial x^b} \quad (1.70)$$

$$\lambda_{a'} = X_{a'}^b \lambda_b \quad (1.71)$$

$$\mathbf{e}_j = U_j^{i'} \mathbf{e}_{i'} \quad (1.44)$$

$$g^{ab} g_{bc} = \delta_c^a = \delta_a^c = g_{ab} g^{bc} \quad (1.78)$$

$$g_{ab} \lambda^a \mu^b = g^{ab} \lambda_a \mu_b = \lambda_a \mu^a = \lambda^a \mu_a \quad (1.79)$$

$$X_c^a = \frac{\partial x^a}{\partial x^c} = \delta_c^a$$

$$X_b^{a'} X_{c'}^{b'} = \delta_{c'}^{a'} = \delta_c^a \quad (1.69)$$

$$X_b^a X_c^{b'} = \delta_c^a \quad (1.68)$$

$$\dot{x}^{a'} = X_{b'}^{a'} \dot{x}^b \quad \text{for a curve } \gamma(t) \text{ where } x = x(t) \text{ and } \dot{x} = \frac{dx}{dt}$$

$$g_{a' b'} = X_{a'}^a X_{b'}^b g_{ab}$$

$$g^{a' b'} = X_a^{a'} X_b^{b'} g^{ab}$$

$$\tau_{b' c'}^{a'} = X_d^{a'} X_{b'}^e X_{c'}^f \tau_{ef}^d \quad (1.75)$$

$$\tau_{b' c'}^{a'} \lambda^{c'} = X_d^{a'} X_{b'}^e \tau_{ef}^d \lambda^f \quad (1.76)$$

$$\tau_{b' c'}^{a'} X_f^{c'} = X_d^{a'} X_{b'}^e \tau_{ef}^d \quad (1.77)$$

$$\tau_{b_1' \dots b_s'}^{a_1' \dots a_r'} = X_{c_1'}^{a_1'} \dots X_{c_r'}^{a_r'} X_{b_1'}^{d_1} \dots X_{b_s'}^{d_s} \tau_{d_1 \dots d_s}^{c_1 \dots c_r} \quad (1.73)$$

$$X_{b' c'}^a \equiv \partial_{b'} X_{c'}^a = \frac{\partial^2 x^a}{\partial x^{b'} \partial x^{c'}} = X_{c' b'}^a \quad (\text{Defn})$$

$$X_{b' c'}^{a'} = X_{b'}^d X_{c'}^{a'} \quad (\text{Th 4.2.4})$$

$$\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}) \quad (2.13)$$

$$\Gamma_{dbc} \equiv \frac{1}{2} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}) \quad (2.33)$$

$$\Gamma_{bc}^a = g^{ad} \Gamma_{dbc} \quad (2.34)$$

$$\Gamma_{abc} = g_{ad} \Gamma_{bc}^d \quad (2.35)$$

$$\partial_c g_{ab} = \Gamma_{abc} + \Gamma_{bac} \quad (2.36)$$

$$\Gamma_{d' e' c'} = X_{d'}^b X_{e'}^f X_{c'}^a \Gamma_{bfa} + X_{e' c'}^a X_{d'}^b g_{ab} \quad (2.32b)$$

$$\Gamma_{b' c'}^{a'} = \Gamma_{fg}^d X_d^{a'} X_{b'}^f X_{c'}^g + X_{c' b'}^d X_d^{a'} \quad (2.32)$$

Field Absolute Derivative Equation (along a curve $\gamma(u)$)

Scalar $\boxed{D\varphi/du = d\varphi/du}$ (2.44)

Contravariant vector $\boxed{D\lambda^a/du = \dot{\lambda}^a + \Gamma_{cd}^a \lambda^c \dot{x}^d}$ (2.45)

Covariant vector $\boxed{D\mu_b/du = \dot{\mu}_b - \Gamma_{bd}^c \mu_c \dot{x}^d}$ (2.46)

Type (2,0) tensor $\boxed{D\tau^{ab}/du \equiv \dot{\tau}^{ab} + [\Gamma_{cd}^a \tau^{cb} + \Gamma_{cd}^b \tau^{ac}] \dot{x}^d}$ (2.48)

Type (0,2) tensor $\boxed{D\tau_{ab}/du \equiv \dot{\tau}_{ab} - [\Gamma_{ad}^c \tau_{cb} - \Gamma_{bd}^c \tau_{ac}] \dot{x}^d}$ (2.49)

Type (1,1) tensor $\boxed{D\tau_b^a/du \equiv \dot{\tau}_b^a + [\Gamma_{cd}^a \tau_b^c - \Gamma_{bd}^c \tau_c^a] \dot{x}^d}$ (2.50)

(r, s) : $\boxed{D\tau_{b_1 \dots b_s}^{a_1 \dots a_r}/du = \dot{\tau}_{b_1 \dots b_s}^{a_1 \dots a_r} + \left[\sum_{k=1}^r \Gamma_{cd}^{a_k} \tau_{b_1 \dots b_s}^{a_1 \dots a_{k-1} c a_{k+1} \dots a_r} - \sum_{k=1}^s \Gamma_{b_k d}^c \tau_{b_1 \dots b_{k-1} c b_{k+1} \dots b_s}^{a_1 \dots a_r} \right] \dot{x}^d}$ (2.51a)

Example: $D\tau_c^{ab}/du = \dot{\tau}_c^{ab} + [\Gamma_{de}^a \tau_c^{db} + \Gamma_{de}^b \tau_c^{ad} - \Gamma_{ce}^d \tau_d^{ab}] \dot{x}^e$

Notation: $\lambda_{,c}^a \equiv \frac{\partial \lambda^a}{\partial x^c}$ $\lambda_{;c}^a \equiv \lambda_{,c}^a + \Gamma_{bc}^a \lambda^b$ (2.53)

Field Covariant Derivative Equation (in a manifold)

Scalar $\boxed{\varphi_{;a} = \partial_a \varphi}$ (2.54)

Contravariant vector $\boxed{\lambda_{;b}^a = \partial_b \lambda^a + \Gamma_{cb}^a \lambda^c}$ (2.55)

Covariant vector $\boxed{\mu_{a;b} = \partial_b \mu_a - \Gamma_{ab}^c \mu_c}$ (2.56)

Type (2,0) tensor $\boxed{\tau^{ab}_{;c} \equiv \partial_c \tau^{ab} + \Gamma_{dc}^a \tau^{db} + \Gamma_{dc}^b \tau^{ad}}$ (2.57)

Type (0,2) tensor $\boxed{\tau_{ab;c} \equiv \partial_c \tau_{ab} - \Gamma_{ac}^d \tau_{db} - \Gamma_{bc}^d \tau_{ad}}$ (2.58)

Type (1,1) tensor $\boxed{\tau_{b;c}^a \equiv \partial_c \tau_b^a + \Gamma_{dc}^a \tau_b^d - \Gamma_{bc}^d \tau_d^a}$ (2.59)

Type (r,s) tensor $\boxed{\tau_{b_1 \dots b_s;c}^{a_1 \dots a_r} = \partial_c \tau_{b_1 \dots b_s}^{a_1 \dots a_r} + \sum_{k=1}^r \Gamma_{dc}^{a_k} \tau_{b_1 \dots b_s}^{a_1 \dots a_{k-1} d a_{k+1} \dots a_r} - \sum_{k=1}^s \Gamma_{b_k c}^d \tau_{b_1 \dots b_{k-1} d b_{k+1} \dots b_s}^{a_1 \dots a_r}}$ (2.59a)

Type (2,1) tensor $\tau^{ab}_{c;d} \equiv \partial_d \tau^{ab}_c + \Gamma_{ed}^a \tau^{eb}_c + \Gamma_{ed}^b \tau^{ea}_c - \Gamma_{cd}^e \tau^{ab}_e$